

STRUCTURAL BIOLOGY

Structure of a fungal 1,3- β -glucan synthaseChao-Ran Zhao^{1†‡§}, Zi-Long You^{1†}, Dan-Dan Chen^{1,2†}, Jing Hang^{2,3,4,5*†}, Zhao-Bin Wang¹, Meng Ji¹, Le-Xuan Wang¹, Peng Zhao¹, Jie Qiao^{2,3,4,5*}, Cai-Hong Yun^{1*}, Lin Bai^{1*}

1,3- β -Glucan serves as the primary component of the fungal cell wall and is produced by 1,3- β -glucan synthase located in the plasma membrane. This synthase is a molecular target for antifungal drugs such as echinocandins and the triterpenoid ibrexafungerp. In this study, we present the cryo-electron microscopy structure of *Saccharomyces cerevisiae* 1,3- β -glucan synthase (Fks1) at 2.47-Å resolution. The structure reveals a central catalytic region adopting a cellulose synthase fold with a cytosolic conserved GT-A-type glycosyltransferase domain and a closed transmembrane channel responsible for glucan transportation. Two extracellular disulfide bonds are found to be crucial for Fks1 enzymatic activity. Through structural comparative analysis with cellulose synthases and structure-guided mutagenesis studies, we gain previously unknown insights into the molecular mechanisms of fungal 1,3- β -glucan synthase.

INTRODUCTION

Fungal diseases are a growing threat to human health. Approximately 1.7 billion people have skin, nail, and hair fungal infections, and more than 1.6 million mortalities occur each year because of serious fungal diseases (1, 2). Furthermore, fungal pathogens cause major losses to agricultural activities and food production and account for approximately 65% of pathogen-driven host extinctions (3, 4). There are currently five classes of clinically approved antifungal drugs: polyenes (amphotericin B), flucytosines, azoles (ketoconazole, itraconazole, fluconazole, voriconazole, and isavuconazole), echinocandins (caspofungin, micafungin, and anidulafungin), and the triterpenoids (ibrexafungerp) (5–7). Among them, the echinocandins and triterpenoids are the only two new classes of antifungal drugs introduced into the market over the past 40 years (1, 5, 6). These two classes inhibit 1,3- β -glucan synthase (GS) in the plasma membrane to block fungal cell wall synthesis (5, 6).

Fungal cell walls are essential for the viability, morphogenesis, and pathogenesis of fungi (8). Fungal cell walls are an attractive target for antifungal drug development because such a cellular structure is absent in human cells (9). Unlike the plant cell wall that is primarily made of cellulose (1,4- β -glucan), the fungal cell wall is composed mainly of 1,3- β -glucan, 1,6- β -glucan, chitin, and mannoproteins (10). Most approved and developing antifungal drugs that target the cell wall function by inhibiting GS (9). This enzyme catalyzes the formation of $\beta(1 \rightarrow 3)$ glycosidic linkages in 1,3- β -glucan by using uridine diphosphate-activated glucose

(UDP-Glc) as the sugar donor and transport the glucan through the membrane (Fig. 1A). Fungal GS is primarily encoded by *FKS1* or *FKS2* (11). *FKS1* is an essential gene in most fungi, including *Candida* and *Aspergillus* species (12). Simultaneous knockout of *Saccharomyces cerevisiae* *FKS1* and *FKS2* is lethal (13). Many echinocandin-resistant mutations have been identified on three hot-spots of *FKS1*, indicating that echinocandins target Fks1 (14, 15). In *S. cerevisiae*, Fks1 is expressed predominantly under normal conditions, whereas Fks2 expression is important during sporulation or growth under particular stressful conditions (13, 16). *S. cerevisiae* Fks1 shares ~88% identity to Fks2 and 56 to 86% identity to *FKSs* in major pathogenic fungi species, suggesting high similar overall structures and mechanisms for different fungal GSs (fig. S1). The GS activity of Fks1 is regulated by Rho1, which is a Ras-like small guanosine triphosphatase (17–21). Deficiency in GS activity was caused by *RHO1* mutants and restored by purified Rho1 in vitro (17, 18, 20).

Fks1 is a GT-A-type glycosyltransferase (GT) that belongs to the GT48 family (<http://www.cazy.org/>). Despite its importance in cell wall synthesis and antifungal drug development, the structure of Fks1 has not been determined, hindering a mechanistic understanding of 1,3- β -glucan synthesis. In this report, the cryo-electron microscopy (cryo-EM) structure of *S. cerevisiae* Fks1 was solved at 2.47-Å resolution, which is the first structure of a GT48 family member. The structure provides insights into the catalytic mechanism and transport of glucan by Fks1 and how echinocandins inhibit the activity of this enzyme.

RESULTS

Purification and characterization of Fks1

To purify Fks1, we used a multicopy plasmid with a strong GAL promoter and a C-terminal triple FLAG tag in *S. cerevisiae*. The protein was isolated using an anti-FLAG affinity column followed by size exclusion chromatography (fig. S2, A and B). Although previous studies have identified Rho1p as a copurified regulatory subunit of the functional glucan synthase (17–20), our purified sample only contained Fks1 (fig. S2B). In agreement with previous findings, our subsequent in vitro UDP-Glo GT assay of the purified Fks1 revealed that Fks1 exhibited minimal activity on its own but

¹Department of Biochemistry and Biophysics, School of Basic Medical Sciences, Peking University, Beijing, China. ²State Key Laboratory of Female Fertility Promotion, Center for Reproductive Medicine, Department of Obstetrics and Gynecology, Peking University Third Hospital, Beijing, China. ³National Clinical Research Center for Obstetrics and Gynecology (Peking University Third Hospital), Beijing, China. ⁴Key Laboratory of Assisted Reproduction, Ministry of Education (Peking University), Beijing, China. ⁵Beijing Key Laboratory of Reproductive Endocrinology and Assisted Reproduction, Beijing, China.

*Corresponding author. Email: hangjbysy@163.com (J.H.); jie.qiao@263.net (J.Q.); yunch@hsc.pku.edu.cn (C.-H.Y.); lbai@pku.edu.cn (L.B.)

†These authors contributed equally to this work.

‡Present address: Department of Otolaryngology Head and Neck Surgery, Beijing Tongren Hospital, Capital Medical University, Beijing, 100730, China.

§Present address: Beijing Key Laboratory of Nasal Diseases, Beijing Institute of Otolaryngology, Beijing, 100005, China.

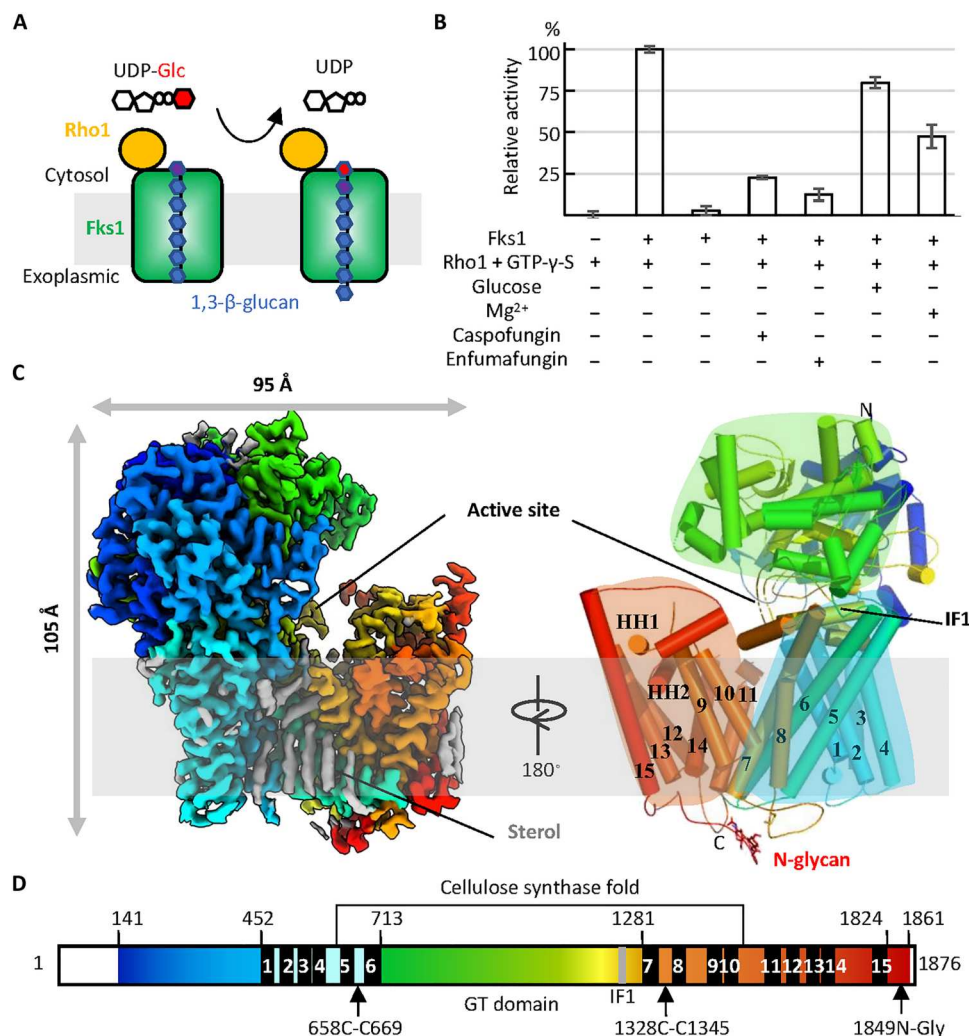


Fig. 1. Enzyme activity and cryo-EM structure of Fks1. (A) Synthesis and transport scheme of 1,3-β-glucan by Fks1. The activity was found to be regulated by Rho1. (B) In vitro UDP-Glc GT assay with purified Fks1 with or without purified Rho1 and GTP-γ-S. The assay under condition with Rho1 and GTP-γ-S was also performed with glucose, Mg²⁺, caspofungin, or enfumafungin. The reaction without Fks1 was used as the control. Data points represent the means ± SD in triplicate. (C) Cryo-EM map and atomic model of Fks1. The polypeptide is shown in rainbow from N to C terminus. (D) Fks1 domain map. Major domains and motifs are labeled. The regions not observed are in hollow.

was highly activated by the addition of purified Rho1 and guanosine 5'-O-(3'-thiotriphosphate) (GTP-γ-S) (Fig. 1B and figs. S2, D and E, and S3, A and B). We also found that the activity of Fks1 was inhibited by the echinocandin drug caspofungin and the triterpenoid enfumafungin, which is the precursor of ibrexgerp (Fig. 1B). Our results showed that the minimal inhibitory concentration of caspofungin and enfumafungin for *S. cerevisiae* was <0.01 and <1 μM, respectively (fig. S3C). The median inhibitory concentration values for caspofungin and enfumafungin in inhibiting Fks1's activity in vitro were ~0.4 and ~5.8 μM, respectively (fig. S3, C to E).

Fks1 belongs to the GT-A superfamily of GTs. While most GT-A fold GTs are metal dependent with a conserved metal coordinate DxD motif (22), Fks1 contains a DAN motif (residues 1102 to 1104) in the corresponding sequence. The DNA motif has been shown to be essential for Fks1 function by a growth complementary assay (fig. S1) (15). To verify whether Fks1 activity is affected by metal ion, we measured the in vitro UDP-Glc hydrolysis activity

of Fks1 in the buffer with or without Mg²⁺ (Fig. 1B). We found that the presence of Mg²⁺ is not required for or even decreased the activity of Fks1, which is consistent with previous studies (11, 23). Besides, the activity of Fks1 was not affected by glucose in the reaction buffer (Fig. 1B), which is also consistent with previous studies that showed that the activity of GS does not need glucose as a starting primer (11, 23).

The overall structure of Fks1

We performed single-particle cryo-EM on purified Fks1 and obtained a cryo-EM three-dimensional (3D) map at 2.47-Å resolution (figs. S2, F and G, and S4). The 3D map exhibited excellent main-chain connectivity and side-chain densities except for some disordered loops (Fig. 1 and fig. S5). We built an atomic model into the map and refined it to good statistics (table S1).

Yeast Fks1 consists of 1876 residues and has a structure measuring of ~105 Å in height and 95 Å in width (Fig. 1, C and D, and fig.

S1). Fks1 can be divided into a transmembrane (TM) region and a large cytosolic region. The TM region contains 15 TM helices that can be grouped into two parts, TM1 to TM8 and TM9 to TM15, based on the packing arrangement of these helices. The TM9 to TM15, along with two amphipathic horizontal helices (HH1 and HH2), tilt away from the first eight TMs, generating a sizable cavity in the middle of the structure facing toward the cytosol. This structural feature likely facilitates the transport of the glucan product through the membrane.

The cytosolic region of Fks1 is composed of two parts: an N-terminal peptide (Met¹⁴¹ to Phe⁴⁵²) and a GT domain inserted between TM6 and TM7 (Ile⁷¹³ to Gln¹²⁸¹) (Figs. 1, C and D, and 2A). The first 140 residues of Fks1 are mostly disordered and invisible in our cryo-EM map. The whole cytosolic region folds into a nine-stranded β sheet, which is sandwiched by two groups of α helices. The cytosolic region is tilted relative to the membrane plane and packs onto the first eight TMs of Fks1, and this interaction is mainly mediated by an amphipathic interface helix (IF1) (Lys¹¹⁷² to Gly¹¹⁸⁸) located at the center. This architecture forms a large horizontal groove corresponding to the enzyme active site, as described below (Fig. 2B).

In addition, we identified several ordered sterol, phospholipid, and detergent molecules in the TM region (Fig. 1, C and D, and figs. S5 and S6). The sterol molecules are found to be around TMs and likely belong to endogenous copurified ergosterol and the detergent cholesteryl hydrogen succinate (CHS) that was

added during purification. Purified Fks1 in buffer without CHS forms aggregates predominantly, suggesting that the sterols help to stabilize the overall Fks1 structure (fig. S2, A to C). One phospholipid molecule is embedded in a cavity surrounded by TM9, TM14, and HH2. Noticeably, its phosphate group directly hydrogen bonds to the side chains of R1455, R1684, and H1688 (fig. S6). This phospholipid probably stabilizes the C-terminal region of Fks1 and plays a structural role.

A structure-based homology search using the online Dali server revealed that the central part of Fks1 (Gly⁶¹⁵ to Asn¹⁴⁹⁰) shares a structural fold with bacterial cellulose synthase BcsA and plant cellulose synthase CesA in the GT-2 family (Figs. 1D, 2, A and C, and 3A and fig. S7) (24–30). Specifically, this part of Fks1 encompasses a cytosolic GT domain and a glucan TM transport domain. Cellulose is a linear D-glucose chain with β 1 to β 4 glycosidic linkages. Both Fks1 and cellulose synthases catalyze glycosidic bond formation using UDP-Glc as the sugar donor and transport the glucan through the membrane. Consequently, we performed structural alignments between Fks1 and BcsA or CesA in complex with substrates (Figs. 2C and 3A and fig. S7) (24, 26) and identified the putative catalytic site and glucan-transporting channel of Fks1. However, no substrate density was observed in the catalytic site or glucan-transporting channel of Fks1, indicating that the Fks1 structure represents an apo form.

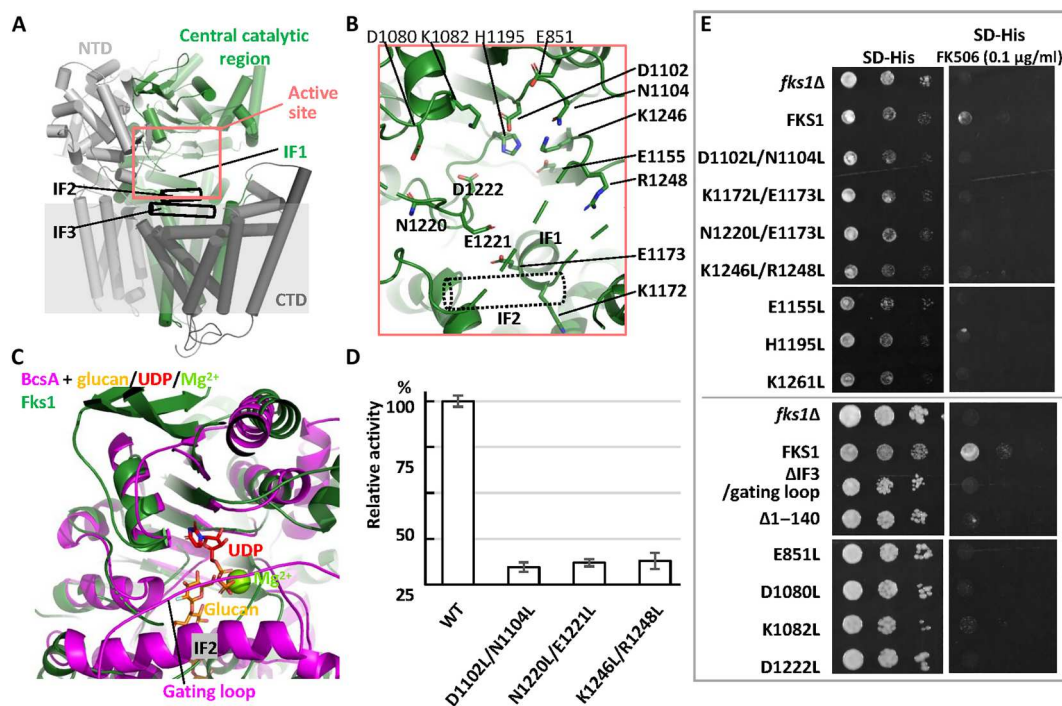


Fig. 2. Putative active site in the GT domain of Fks1. (A) Overall structure of Fks1 in cartoon representation. The central catalytic region (green) is sandwiched by N-terminal (gray) and C-terminal (black) regulatory domains. The putative active site is highlighted by a circle (salmon). Flexible IF2 and IF3 are highlighted by two open black cylinders. (B) A putative active site in Fks1, with residues forming the pocket shown in stick representation. (C) Structural comparison between the soluble GT domains of Fks1 (green) and BcsA (magenta) in complex with cellulose (orange), UDP (red), and Mg²⁺ (lemon). (D) UDP-Glc hydrolysis activity of wild-type (WT) Fks1 and Fks1 mutant. Data points represent the means $\pm$ SD in triplicate. (E) Growth complementation of *fks1* Δ cells with empty plasmid (*fks1* Δ) or plasmid carrying either WT FKS1 (FKS1) or mutants. Either 2 μ l (top box) or 5 μ l (bottom box) of cells was spotted onto SD-His plates and SD-His plates with FK506 (0.1 μ g/ml). Plates were incubated at 30°C for 2 days.

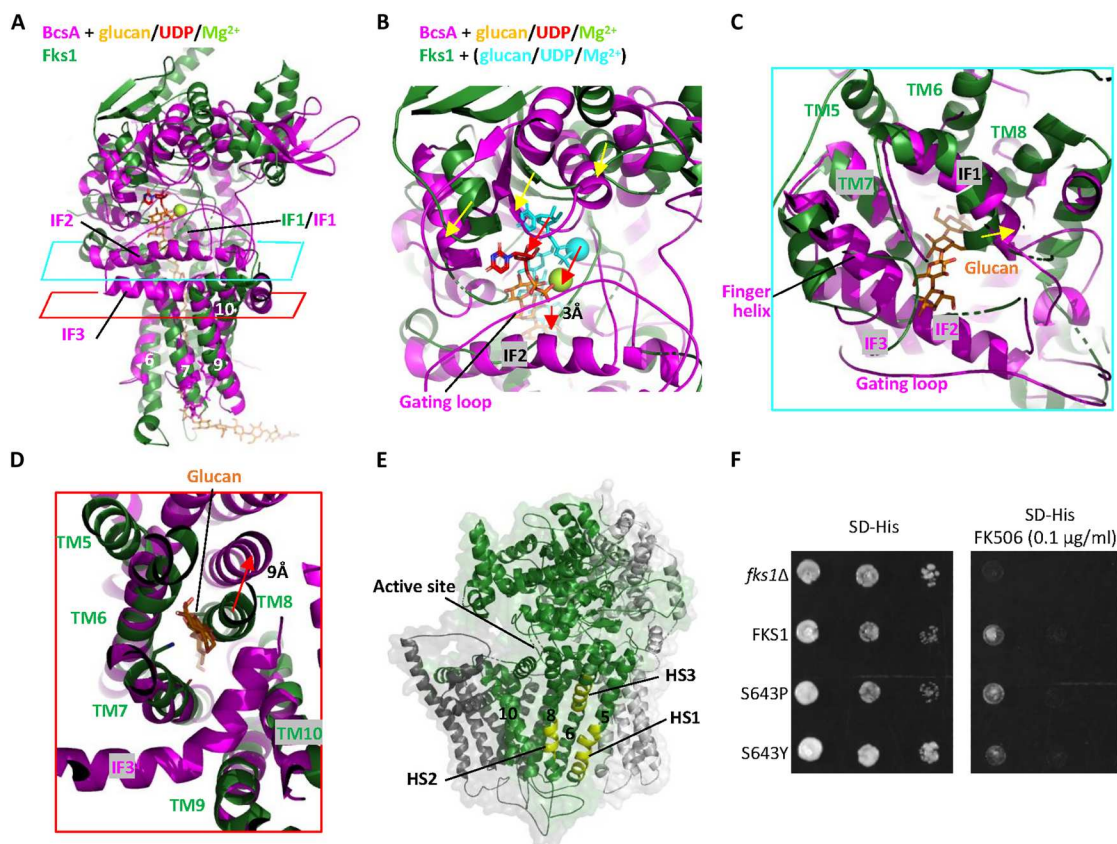


Fig. 3. Putative glucan-transporting channel of Fks1. (A) Structural comparison between the central catalytic region of Fks1 (green) and BcsA (Protein Data Bank ID: 5EJZ, magenta) in complex with cellulose (orange), UDP (red), and Mg^{2+} (lemon) by aligning their respective TMs. The cyan and red rhomboids mark the positions for cut-in views shown in (C) and (D). (B) A close-up view of the cytosolic region in (A). Substrates in cyan are modeled onto Fks1's active site basing on structural alignment of GT domains of Fks1 and BcsA in Fig. 2B. Movements of the GT domain and substrates are highlighted by yellow and red arrows, respectively. The glucan shifts downward by ~ 3 Å. (C and D) Cut-in views from the cytosolic side of the superposition of Fks1 and BcsA in (A). TM8 and IF1 of Fks1 form part of the putative glucan transport path. Movements of the TM8 and IF1 are highlighted by red and yellow arrows, respectively. Compared with BcsA, TM8 of Fks1 is shifted inward by ~ 9 Å and occupy the glucan transporting path, resulting in a closed channel. Corresponding sequences of IF2 and IF3 and gating loop are disordered in Fks1. (E) Mapping of the echinocandins resistance mutant hotspots (HS1 to HS3) onto the structure of Fks1. (F) Growth complementation of *fks1Δ* cells with empty plasmid (*fks1Δ*) or plasmid carrying either WT Fks1 (FKS1) or mutants (S643P and S643Y in HS2). Cells were serially diluted, spotted onto SD-His plates and SD-His plates with FK506 (0.1 $\mu\text{g/ml}$), and incubated at 30°C for 2 days.

The putative catalytic site of Fks1

Previous studies have demonstrated the GT-A fold adopts a conserved $\alpha/\beta/\alpha$ sandwich with active site located at the center (31, 32). Similar to other GT-A fold proteins, the cytosolic GT domain of Fks1 consists of a central nine-stranded β sheet surrounded by two groups of α helices. The putative active site of Fks1 was proposed to be around the end of central β sheet, adjacent to the TM region. Notably, this putative active site comprises several charged residues, including E851, D1080, K1082, D1102, E1155, K1172, E1173, E1221, D1222, K1246, and R1248 (Fig. 2B). Among them, D1102 and D1222 correspond to the second and third aspartic acids of the conserved D, D, D signature found in cellulose synthases (D180, D246, and D343 in BcsA). In the structure of BcsA, D246 in a DxD motif coordinates the divalent cation, whereas D343 in a TED motif of the finger helix is responsible for substrate binding and glucose polymer elongation (24, 25, 29). To reveal function of these key residues in detail, we superimposed the GT domain of Fks1 with the GT domain of bacterial BcsA and *Bacillus subtilis* SpsA (Fig. 2C and fig. S8A) (24, 33). We determined

their root mean square deviation (RMSD) to be 2.4 and 2.7 Å, respectively, indicating a high conservation of the GT domain of Fks1. SpsA is a GT that transfers the glycosyl moiety from an activated UDP-sugar donor to a specific acceptor and was implicated in the synthesis of the spore coat of *B. subtilis*. By investigating the substrates positions in the BcsA and SpsA structures and the active site of Fks1, we propose that E851, D1080, K1082, D1102, and N1104 at the top of the active site of Fks1 stabilize the sugar donor UDP-Glc, H1195, E1221, D1222, and K1246 and probably form the acceptor glucose binding site, catalyze the sugar transfer, and participate in glucan elongation, and K1172 and E1173 in IF1, E1155, and R1248 may stabilize the glucan product in Fks1 (fig. S8, A and B). All these residues in the active site are conserved among fungal GSs (fig. S1).

To evaluate the structure-function relationship of the aforementioned key residues, we introduced mutations and subsequently purified the mutant proteins for in vitro UDP-Glo GT assay. We found that the enzymatic activities of D1102L/N1104L, N1220L/E1221L, and K1246/R1248L mutants were markedly reduced when

compared to the activity of the wild-type (WT) enzyme (Fig. 2D). Furthermore, we performed *in vivo* growth complementation assays of the *fks1* mutants using the yeast *fks1Δ* strain. Previous studies indicated that the *fks1Δ* strain is hypersensitive to the immunosuppressant FK506 (11), and this deficiency can be partially complemented by a plasmid carrying the WT *fks1* gene. We found that E851L, D1080L, K1082L, D1221L, D1102L/N1104L, E1155L, K1172L/E1173L, H1195L, N1220L/E1221L, and K1246/R1248L mutants disrupted Fks1 function substantially as they were unable to rescue *fks1Δ* yeast growth (Fig. 2E).

The central catalytic region of our Fks1 structure in the apo state is superimposable with the structure of BcsA for both the GT domain and the TM glucan-transporting domain (Figs. 2C and 3, A to D). However, the GT domain of Fks1 is shifted upward and away by around 3 to 8 Å from the membrane region relative to that of BcsA (Fig. 3B). This structural feature leads to a more open active site, which probably facilitates substrate binding. We hypothesize that the GT domain will move downward upon substrate binding and thus push the glucan product out. To further confirm this hypothesis, we analyzed the cryo-EM structure of Fks1 using 3D variability analysis (3DVA), which is a newly developed software in CryoSPARC to reveal the continuous variability and discrete heterogeneity of a structure (34). We found that the cytosolic domain indeed moves up and down relative to the TM region in two extremes, and accordingly, the substrate binding site becomes more open and closed (movie S1). We also noticed that a motif in GT domain (R862-V870 motif) interacts a cytosolic loop between TMH15 and HH1 in the C-terminal domain (CTD) (TMH15-HH1 loop) when the cytosolic domain moves down (movie S1 and fig. S6A). The CTD may regulate the activity of Fks1 through this interaction. Deletion of the R862-V870 motif was found to disrupt the function of Fks1 in the UDP-Glo GT assay and growth complementation assay, indicating its importance (fig. S6, D and E).

Furthermore, we found that despite an RMSD of ~2.0 Å, the predicted Fks1 model by AlphaFold2 (35, 36) is more compact than our cryo-EM structure, with the cytosolic domain and CTD undergoing a rigid-body movement toward each other, forming a more closed active site (fig. S9, A to C). This compact conformation probably represents the substrate bound conformation of Fks1. According to the predicted Fks1-Rho1 complex model by AlphaFold2, Rho1 binds to Fks1 through interactions with both the R862-V870 motif and the TMH15-HH1 loop (fig. S9D). The interface is characterized by several ionic bonds between E69, D70, and E102 of Rho1 and R858, R862, and K1788 of Fks1. The binding of Rho1 in the predicted complex model seems to help Fks1, improving the stability of the R862-V870 motif and regulating the movement of GTD and CTD.

The putative TM glucan transporting channel

The putative TM glucan-transporting channel of Fks1 is proposed to be positioned directly below the active site and composed of TM5 to TM10 (Fig. 3, A to D, and fig. S10A). Unlike the channel accommodating the translocating glucan in the BcsA and CesA structures (24, 26), there is no glucan observed in our structure of Fks1, indicating the structure to be in the apo state. Furthermore, the putative glucan-transporting channel is only accessible from cytosolic side but is closed in the extraplasmic side. Structural alignment between TM5 to TM10 of Fks1 and the corresponding region of

BcsA and CesA8 revealed that TM5 to TM7 and TM9 and TM10 are well aligned with an RMSD of ~3.0 Å, while TM8 of Fks1 is shifted inward and occupies the glucan transporting path, resulting in a closed channel (Fig. 3D and fig. S7). Furthermore, compared with the ordered cytosolic entry of the channel in BcsA and CesA, which is mainly formed by three horizontal interface helices (IF1 to IF3), a TM helix (TM5 in BcsA and TM2 in CesA), and a conserved finger helix and gating loop, the corresponding sequences of IF2 and IF3 and the gating loop in Fks1 are flexible, and IF1 moves toward the channel by ~5 Å (Fig. 3, C and D). These structural features of Fks1 are likely caused by the absence of the glucan substrate. The AlphaFold2 predicted model of Fks1 showed IF2 and IF3 fold right under active site like in CesA structure and may represent the substrate bound conformation of Fks1 (fig. S9). Although the flexible regions are not conserved between Fks1 and cellulose synthase, we found that they are still essential for Fks1's function according to the disruption of the K1261L mutation in IF2 and the IF3/gating loop deletion to Fks1 function in the growth complementation assay (Fig. 2C). Together, we propose that TM8 and IF1 need to move outward upon glucan synthesis to open the channel for glucan translocation, and IF2 and IF3 play important roles during this process.

Crossing the membrane, the putative glucan-transporting channel featured with many hydrophobic residues and few polar residues (such as D684, N1300, N1301, S1307, T1314, T1359, and S1361) (fig. S10, A and B). To investigate the functional significance of the polar residues, we introduced three mutations (N1300F/N1301F, T1314F, and S1361F) and performed complementation assays on *fks1Δ* cells. We found that all three mutations disrupted Fks1 function *in vivo*, as they were unable to rescue the *fks1Δ* yeast growth than the plasmid carrying the WT *FKS1* (fig. S11C). Because the glucan is negatively charged, we suggest that the polar residues in the channel may assist the glucan transporting.

Structural interpretation of echinocandin resistance

Previously reported echinocandin-resistance mutations are mainly distributed in three "hotspot" regions (HS1: 635 to 649, HS2: 1354 to 1361, and HS3: 690 to 700). We identified that these regions belong to TM5, TM8, and TM6 of Fks1, respectively, and are gathered into a groove within the TM region, which is likely the putative drug-binding site (Fig. 3E). Furthermore, these TM helices play crucial roles in the proposed glucan-transporting channel and are distal from the active site. Specifically, HS1 and HS2 interact with each other, and this interface is expected to undergo conformational changes if TM8 shifts outward during glucan transport, as proposed above. Therefore, echinocandins likely inhibit Fks1 activity by affecting glucan translocation. We found that two frequent echinocandin mutants, S643P and S643Y, located in HS2, maintained native Fks1 function in the growth complementation assay (Fig. 3F). The *in vitro* activity assay showed that the enzyme activity of S643P was ~130% of WT Fks1, and it was only slightly inhibited by caspofungin but notably inhibited by enfumafungin (fig. S3F). Therefore, these mutants escape the inhibition of echinocandins either by disrupting the drug binding or improving catalytic efficiency to overcome the inhibition.

Essential role of two extracellular disulfide bonds of Fks1

The extracellular side of Fks1 only contains a few loops, but two disulfide bonds (Cys⁶⁵⁸ with Cys⁶⁶⁹ and Cys¹³²⁸ with Cys¹³⁴⁵) were

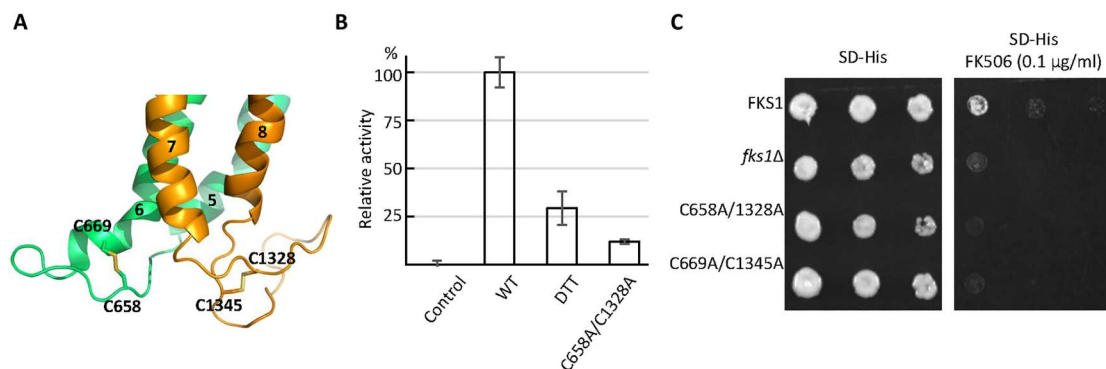


Fig. 4. The two extracellular disulfide bonds of Fks1. (A) Two disulfide bonds (Cys⁶⁵⁸ with Cys⁶⁶⁹ and Cys¹³²⁸ with Cys¹³⁴⁵) are located in the exoplasmic loops between TM5 and TM6 (EL3) and TM7 and TM8 (EL4) of Fks1, respectively. (B) UDP-Glc hydrolysis activity of WT Fks1 (WT), Fks1 in a buffer containing DTT, and Fks1 C658A/C1328A mutant. The reaction without Fks1 was used as the control. Data points represent the means $\pm$ SD in triplicate. (C) Growth complementation of *fks1* Δ cells with a plasmid carrying Fks1 mutant (C658A/C1328A or C669A/C1345A). The empty plasmid (*fks1* Δ) and a plasmid carrying WT *FKS1* (FKS1) were used as controls.

found in the exoplasmic loops between TM5 and TM6 (EL3) and TM7 and TM8 (EL4) of Fks1, respectively (Fig. 4A). The two disulfide bonds are very close to neighboring TM helices and appear to shorten EL3 and EL4. Given that TM8 may shift markedly to open the channel during glucan transport as proposed above, the limited relative flexibility of TM5 to TM8 by disulfide bonds may be important for regulating Fks1 activity.

We first performed the *in vitro* UDP-Glc GT assay to investigate the function of these disulfide bonds and found that Fks1 activity is reduced notably by the addition of dithiothreitol (DTT), a disulfide bond reducing agent (Fig. 4B). Furthermore, we generated Fks1 double mutants (C658A/C1328A), and the activity assays showed that C658A/C1328A mutant had reduced activities by $\sim$ 90% when compared with the activity of WT Fks1 (Fig. 4B and fig. S3E). We also performed *in vivo* growth complementation assays of two Fks1 double mutants (C658A/C1328A and C669A/C1345A) using the yeast *fks1* Δ strain. We found that neither C658A/C1328A nor C669A/C1345A mutants were able to rescue *fks1* Δ yeast growth on plates containing FK506 (0.1 μ g/ml), supporting the essential functional role of the two disulfide bonds (Fig. 4C). In addition, an N-glycan on Asn¹⁸⁴⁹ was found to locate directly above the disulfide bond formed between Cys¹³²⁸ and Cys¹³⁴⁵ and likely contributes to maintain the stability of the structure (Fig. 1C and fig. S5).

DISCUSSION

As key enzymes in cell wall construction, Fks1 and cellulose synthases synthesize 1,3- β -glucan and 1,4- β -glucan (cellulose), respectively. Despite the low sequence identity of 11 to 13%, the core domains of Fks1 and cellulose synthases from bacteria and plants share a conserved fold containing a cytosolic GT domain and a TM glucan-transporting domain. This structural and functional similarities indicate that Fks1 functions by using a similar mechanism as cellulose synthases. Our structure of Fks1 was determined in the apo state, while previous high-resolution structures of cellulose synthases were all reported in complex with substrate. Therefore, we proposed a working model of Fks1 based on the analysis of our structure and structural comparison with cellulose synthases (Fig. 5). In the apo state of Fks1, the putative active site is open for substrate binding, whereas the putative glucan-transporting

channel is closed and occupied by TM8. Flexible IF2 and IF3 probably also facilitate substrate entering the active site. We further propose that substrate binding and the glycosyl transfer reaction trigger the movement of the GT domain toward the glucan-transporting channel and thus lead to channel opening and glucan elongation. This working model is also supported by the Fks1 structure predicted by AlphaFold2 and the 3DVA analysis of our cryo-EM structure of Fks1 as mentioned above.

A recent study reported a preliminary hexameric structure of the putative *Candida glabrata* GS complex at $\sim$ 14-Å resolution using cryo-electron tomography (cryo-ET) and subtomogram analysis (fig. S11A) (37). To investigate the mechanism of Fks1 *in vivo*, we compared our high-resolution Fks1 structure with the cryo-ET map of the hexameric GS (fig. S11, A and B). Unexpectedly, we found that our Fks1 structure is different from one subunit of the cryo-ET map in terms of either size or features. Specifically, the TM region of our structure is much larger than one subunit of the cryo-ET map, while the cytosolic region of our structure is much shorter. In the cryo-ET study, the authors proposed that each subunit of the putative GS complexes contains two notable cytosolic domains, the N-terminal and central catalytic domains. However, our structure of Fks1 only contains one large cytosolic GT domain. Moreover, we found that the cryo-ET map of putative hexameric GS is well superposed with the structure of the hexameric Pma1, which is the most abundant protein in the plasma membrane of most fungi including *C. glabrata* (fig. S11C) (38). Notably, the cytosolic domains of Pma1 are well consistent with features of the cryo-ET map. Assignment of particles in low-resolution cryo-ET is always challenging. Although the authors compared native and Fks1-overexpressing strains to prove the identity of particles to be GS in that cryo-ET study, the evidence was not direct and was easily affected by bias. The possibility that the most abundant hexameric Pma1 in plasma membrane was mistaken for the GS cannot be ruled out. Therefore, whether the reported cryo-ET structure of putative hexameric GS is corresponding to GS or Pma1 still needs more studies to confirm. According to our results, we tend to believe that Fks1 functions as a monomer rather than a hexamer.

During preparation of this manuscript, another group reported the cryo-EM structures of *S. cerevisiae* *FKS1* and the echinocandin-resistant mutant *FKS1*(S643P) (fig. S12) (39). The structures and our structure are superimposable with an RMSD of $\sim$ 1.7 Å,

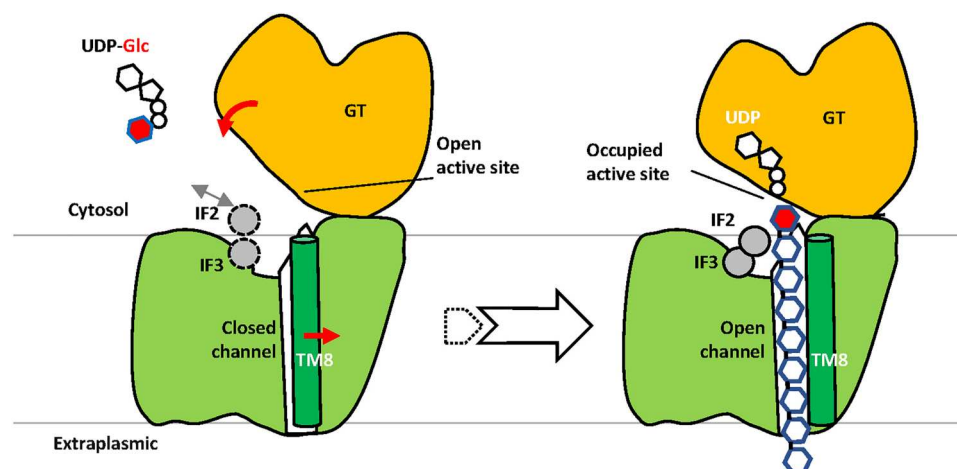


Fig. 5. Proposed catalytic mechanism of Fks1. The cryo-EM structure reflects the apo state (left) with flexible IF2 and IF3 and an empty open, active site for substrate binding. The glucan transporting channel is closed and engaged by TM8. Substrate binding and subsequent reaction may trigger the downward movement of the GT domain, which pushes the glucan product out of the cell. IF2 and IF3 may stabilize the glucan during glucan elongation, and TM8 is shifted outward, resulting in an open channel. The conformation of Fks1 in complex with substrates (right) is modeled according to the structure of cellulose synthases in complex with UDP and cellulose. More details are provided in the main text.

indicating their high similarity (fig. S13). Both works identified the cellulose synthase-like fold as the core structure of Fks1 and proposed similar catalytic mechanism and the echinocandin-resistance mechanism. The major substantial difference observed in structural comparison is notable conformational change of several key elements around the active site, including IF1, part of TM8, and a helix linker between TM8 and IF3 (TM8-IF3 linker). In our structure, IF1 and TM8-IF3 linker are all extending inside, in a similar position as these elements in cellulose synthase, chitin synthase, and hyaluronan synthase (26, 40, 41). In contrast, IF1 and TM8-IF3 linker in published Fks1 models are extending outside, and part of TM8 is shifted by ~ 12 Å. These differences are probably caused by the different detergent used in two works [lauryl maltose neopentyl glycol (LMNG)-CHS in our work; Glyco-diosgenin in published work]. However, it also indicates that TM8 is moveable in different conditions, which supports our proposed working model of Fks1. Besides, the bound lipids were mainly identified to be ordered sterol molecules in our structure, which is different from lipid alkyl tails proposed in the published paper. The lipid densities around the echinocandin resistant mutant site S643 in our structure are relatively weak (fig. S12).

In conclusion, our cryo-EM structure and structure-based mutagenesis analysis of Fks1 have provided deep insights into the molecular mechanisms for GT activity and TM glucan transport of a fungal GS and echinocandin resistance by Fks1 mutants. Meanwhile, one apo structure of Fks1 is not enough to fully understand the detailed mechanisms of Fks1, including its catalytic cycle, activation by Rho1, and inhibition by echinocandins. Capturing structures of Fks1 in complex with substrates and inhibitors will help to fully unveil these mechanisms. Besides, the invisible N-terminal loop of Fks1 in our structure is also probably essential as we found that the deletion of first 140 residues strongly decreased Fks1's function in the growth complementation assay (Fig. 2E). How this loop works also awaits further studies. Given the critical role of GS in pathogenic fungal infections, our work provides a platform for developing new small antifungal molecules.

MATERIALS AND METHODS

Expression and purification of Fks1

We tagged *FKS1* with a C-terminal triple FLAG to the modified pRS423 vector. The yeast strain BY4742 was transformed and cultured by synthetic histidine-dropout medium (SD-His) for about 20 hours and then transferred to YPG medium (10 g of yeast extract, 20 g of peptone, and 20 g of D-galactose per liter) for 12 hours before harvest. Cells were resuspended in lysis buffer [20 mM Tris-HCl (pH 7.4), 0.2 M sorbitol, 50 mM potassium acetate, 2 mM EDTA, and 1 mM phenylmethylsulfonylfluoride (PMSF)] and then lysed using a French press at 15,000 psi. We centrifuged the lysate at 10,000g for 30 min at 4°C and collected the supernatant for another centrifuge cycle at 100,000g for 60 min at 4°C. The membrane pellet was collected and then resuspended in buffer A containing 10% glycerol, 20 mM Tris-HCl (pH 7.4), 1% n-Dodecyl-β-D-Maltopyranoside, 0.1% CHS, 500 mM NaCl, 1 mM MgCl₂, 1 mM EDTA, and 1 mM PMSF. After incubation for 30 min at 4°C, the mixture was centrifuged for 30 min at 100,000g to remove the insoluble membrane. We loaded the supernatant into a pre-equilibrated anti-FLAG (M2) affinity column (GenScript) at 4°C and washed the affinity gel with buffer B [20 mM Hepes (pH 7.4), 150 mM NaCl, 0.01% LMNG, 0.001% CHS, and 1 mM MgCl₂]. The proteins were eluted with buffer B containing 3× FLAG peptide (0.15 mg/ml) and were further purified in a Superose 6 10/300 Increase gel filtration column in buffer C [20 mM Hepes (pH 7.4), 150 mM NaCl, 0.003% LMNG, 0.0003% CHS, and 1 mM MgCl₂]. The purified proteins were assessed by SDS-polyacrylamide gel electrophoresis gel and concentrated for cryo-EM analysis. To maintain consistency with previous functional assays (11, 17, 18), the samples used for UDP-Glc GT assay were prepared using the same procedure as described above except that CHAPS and CHS were used to solubilize the membrane and stabilize the membrane protein.

Expression and purification of Rho1

RHO1 was cloned into pET-28a vector with a C-terminal His tag and expressed in *Escherichia coli* BL21(DE3). Bacteria were grown

in LB broth at 37°C until the optical density at 600 nm (OD_{600}) reached approximately 0.6, and then, the cells were induced with 1 mM isopropyl- β -D-1-thiogalactopyranoside at 25°C overnight. The cell pellet was collected and resuspended in 50 ml of buffer [20 mM Hepes (pH 7.4), 150 mM NaCl, 1 mM $MgCl_2$, 1 mM $MnCl_2$, and 1 mM EDTA] supplemented with protease inhibitors (1 mg/ml each of deoxyribonuclease, pepstatin, leupeptin, and aprotinin and 1 mM PMSF). Cells were lysed on ice by sonication. Pellets were removed by centrifugation for 30 min at 12,000 rpm, and the supernatant was loaded onto a 2-ml column of nickel-nitrilotriacetic acid agarose. The column was washed with 20 mM Hepes (pH 7.4), 20 mM imidazole, and 150 mM NaCl, and Rho1 was eluted with wash buffer supplemented with 200 mM imidazole. The protein eluate was concentrated and further purified by size exclusion chromatography on a Superose 6 10/300 GL column (GE Healthcare). Peak fractions were collected for functional assay.

Cryo-electron microscopy

Aliquots (2.5 μ l) of Fks1 at a concentration of about 3 mg/ml were placed on glow-discharged holey carbon grids (Quantifoil Au R1.2/1.3, 300 mesh) and were flash frozen in liquid ethane using an FEI Vitrobot Mark IV. Grids were screened in a 300-keV FEI TF30 electron microscope from School of Basic Medical Sciences, Peking University. Cryo-EM data were collected automatically with Serial-EM in a 300-kV FEI Titan Krios electron microscope from School of Physics, Peking University and Institute of Biophysics Chinese Academy of Sciences. We collected cryo-EM data with defocus values ranging from -1.0 to 2.0 μ m. The microscope was operated with a K3 direct detector at a nominal magnification of $\times 130,000$. The total doses were 50 to 60 electrons per square angstrom at the sample level.

Cryo-EM image processing

We used the programs MotionCorr-2.0 (42) for motion correction, and CTFFIND-4.1 (43) for calculation of contrast transfer function parameters. We used CryoSPARC for all remaining steps (particle picking and extraction, 2 cycles of 2D classification, 1 cycle of ab initio reconstruction, 1 cycle of heterogeneous refinement, and 1 cycle of nonuniform refinement) (34, 44). The resolution of the map was estimated by the gold-standard Fourier shell correlation at a correlation cutoff value of 0.143.

We totally collected 2778 raw movie micrographs. A total of 3,760,856 particles were picked automatically for 2D and 3D classifications. On the basis of the quality of the three 3D classes, 512,915 particles were retained for further refinement, resulting in a 2.47-Å average resolution 3D map using C1 symmetry.

Structural modeling, refinement, and validation

We used the predicted Fks1 structure by Alphafold2 as initial model (35, 36), fitted it into our map, and corrected it in Coot (45) and Chimera (46). The complete Fks1 model was refined by real-space refinement in the Phenix program (47) and subsequently adjusted manually in Coot. Last, the model was validated using MolProbity (48). Structural figures were prepared in Chimera and PyMOL (<https://pymol.org/2/>).

UDP-Glo GT assay

The activity of WT Fks1 were measured using the UDP-Glo GT assay from Promega (catalog no. V6961), which can measure the

activity of any GT that uses a UDP-sugar as a substrate. Each reaction contained 8 ng of purified Fks1, 20 mM Hepes (pH 7.4), 150 mM NaCl, 0.6% CHAPS, 0.06% CHS, 10% glycerol, and 1 mM EDTA in a total volume of 5 μ l. For other reaction conditions, 0.05 μ g of purified Rho1, 1 mM GTP- γ -S, 2 mM glucose, 2 mM Mg^{2+} , 1 nM to 1 mM caspofungin, or 1 nM to 1 mM enfumafungin was also added. The mixture was first incubated for 30 min at 30°C, and then, 5 μ l of UDP detection reagent was added. The mixture was incubated for 60 min at room temperature, and then, luminescence was detected by Synergy H1 Hybrid Multi-Mode Microplate Reader (BioTek).

Broth microdilution assay

Wide-type yeast strain was first cultured in Synthetic Drop-Out (SD) medium at 30°C overnight as the stock. Then, cells were diluted and same amount of cells in different tubes were further cultured in SD medium containing 0.1 nM to 1 mM caspofungin or enfumafungin at 30°C for 12 hours. Then, OD_{600} was detected by Synergy H1 Hybrid Multi-Mode 414 Microplate Reader (BioTek).

Colony growth assay

Wide-type yeast strain and the *FKS1* knockout strains were first grown in SD medium at 30°C overnight to the same OD. Then, 1:10 serial dilutions of the cells were spotted onto SD plates with or without FK506 (0.1 μ g/ml), incubated at 30°C for 2 days, and examined for growth.

Supplementary Materials

This PDF file includes:

Figs. S1 to S13

Table S1

Legend for movie S1

Other Supplementary Material for this manuscript includes the following:

Movie S1

REFERENCES AND NOTES

1. G. D. Brown, D. W. Denning, N. A. R. Gow, S. M. Levitz, M. G. Netea, T. C. White, Hidden killers: Human fungal infections. *Sci. Transl. Med.* **4**, 165rv13 (2012).
2. F. Almeida, M. L. Rodrigues, C. Coelho, The still underestimated problem of fungal diseases worldwide. *Front. Microbiol.* **10**, 214 (2019).
3. M. C. Fisher, D. A. Henk, C. J. Briggs, J. S. Brownstein, L. C. Madoff, S. L. McCraw, S. J. Gurr, Emerging fungal threats to animal, plant and ecosystem health. *Nature* **484**, 186–194 (2012).
4. S. Savary, A. Ficke, J.-N. Aubertot, C. Hollier, Crop losses due to diseases and their implications for global food production losses and food security. *Food Secur.* **4**, 519–537 (2012).
5. D. W. Denning, Echinocandin antifungal drugs. *Lancet* **362**, 1142–1151 (2003).
6. M. W. McCarthy, Pharmacokinetics and pharmacodynamics of ibrexafungerp. *Drugs R&D* **22**, 9–13 (2022).
7. F. C. Odds, A. J. P. Brown, N. A. R. Gow, Antifungal agents: Mechanisms of action. *Trends Microbiol.* **11**, 272–279 (2003).
8. N. A. R. Gow, J.-P. Latge, C. A. Munro, The fungal cell wall: Structure, biosynthesis, and function. *Microbiol. Spectr.* **5**, 10.1128/microbiolspec.FUNK-0035-2016, (2017).
9. S. L. Lima, A. L. Colombo, J. N. de Almeida Junior, Fungal cell wall: Emerging antifungals and drug resistance. *Front. Microbiol.* **10**, 2573 (2019).
10. G. Lesage, H. Bussey, Cell wall assembly in *Saccharomyces cerevisiae*. *Microbiol. Mol. Biol. Rev.* **70**, 317–343 (2006).
11. C. M. Douglas, F. Foor, J. A. Marrinan, N. Morin, J. B. Nielsen, A. M. Dahl, P. Mazur, W. Banginsky, W. Li, M. el-Sherbeini, The *Saccharomyces cerevisiae* FKS1 (ETG1) gene encodes an

- integral membrane protein which is a subunit of 1,3- β -D-glucan synthase. *Proc. Natl. Acad. Sci. U.S.A.* **91**, 12907–12911 (1994).
12. K. R. Healey, P. Paderu, X. Hou, C. Jimenez Ortigosa, N. Bagley, B. Patel, Y. Zhao, D. S. Perlin, Differential regulation of echinocandin targets Fks1 and Fks2 in *Candida glabrata* by the post-transcriptional regulator Ssd1. *J. Fungi (Basel)* **6**, 143 (2020).
 13. P. Mazur, M. Morin, W. Baginsky, M. el-Sherbeini, J. A. Clemas, J. B. Nielsen, F. Foor, Differential expression and function of two homologous subunits of yeast 1,3- β -D-glucan synthase. *Mol. Cell. Biol.* **15**, 5671–5681 (1995).
 14. D. S. Perlin, Resistance to echinocandin-class antifungal drugs. *Drug Resist. Updat.* **10**, 121–130 (2007).
 15. M. E. Johnson, T. D. Edlind, Topological and mutational analysis of *Saccharomyces cerevisiae* Fks1. *Eukaryot. Cell* **11**, 952–960 (2012).
 16. M. Gomar-Alba, M. A. Morcillo-Parra, M. L. Del Olmo, Response of yeast cells to high glucose involves molecular and physiological differences when compared to other osmostress conditions. *FEMS Yeast Res.* **15**, fov039 (2015).
 17. H. Qadota, C. P. Python, S. B. Inoue, M. Arisawa, Y. Anraku, Y. Zheng, T. Watanabe, D. E. Levin, Y. Ohya, Identification of yeast Rho1p GTPase as a regulatory subunit of 1,3- β -glucan synthase. *Science* **272**, 279–281 (1996).
 18. M. Arellano, A. Duran, P. Perez, Rho1 GTPase activates the (1,3)- β -glucan synthase and is involved in *Schizosaccharomyces pombe* morphogenesis. *EMBO J.* **15**, 4584–4591 (1996).
 19. M. Diaz, Y. Sanchez, T. Bennett, C. R. Sun, C. Godoy, F. Tamanoi, A. Duran, P. Perez, The *Schizosaccharomyces pombe* cwg2+ gene codes for the beta subunit of a geranylgeranyltransferase type I required for β -glucan synthesis. *EMBO J.* **12**, 5245–5254 (1993).
 20. P. Mazur, W. Baginsky, In vitro activity of 1,3- β -D-glucan synthase requires the GTP-binding protein Rho1. *J. Biol. Chem.* **271**, 14604–14609 (1996).
 21. J. Drgonová, T. Dragon, K. Tanaka, R. Kollár, G. C. Chen, R. A. Ford, C. S. Chan, Y. Takai, E. Cabib, Rho1p, a yeast protein at the interface between cell polarization and morphogenesis. *Science* **272**, 277–279 (1996).
 22. C. A. Wiggins, S. Munro, Activity of the yeast MNN1 α -1,3-mannosyltransferase requires a motif conserved in many other families of glycosyltransferases. *Proc. Natl. Acad. Sci. U.S.A.* **95**, 7945–7950 (1998).
 23. E. M. Shematek, J. A. Braatz, E. Cabib, Biosynthesis of the yeast cell wall. I. Preparation and properties of β -(1 leads to 3)glucan synthetase. *J. Biol. Chem.* **255**, 888–894 (1980).
 24. J. L. W. Morgan, J. Strumillo, J. Zimmer, Crystallographic snapshot of cellulose synthesis and membrane translocation. *Nature* **493**, 181–186 (2013).
 25. J. L. W. Morgan, J. T. McNamara, J. Zimmer, Mechanism of activation of bacterial cellulose synthase by cyclic di-GMP. *Nat. Struct. Mol. Biol.* **21**, 489–496 (2014).
 26. P. Purushotham, R. Ho, J. Zimmer, Architecture of a catalytically active homotrimeric plant cellulose synthase complex. *Science* **369**, 1089–1094 (2020).
 27. X. Zhang, Y. Xue, Z. Guan, C. Zhou, Y. Nie, S. Men, Q. Wang, C. Shen, D. Zhang, S. Jin, L. Tu, P. Yin, X. Zhang, Structural insights into homotrimeric assembly of cellulose synthase CesA7 from *Gossypium hirsutum*. *Plant Biotechnol. J.* **19**, 1579–1587 (2021).
 28. L. Holm, L. M. Laakso, Dali server update. *Nucleic Acids Res.* **44**, W351–W355 (2016).
 29. J. L. W. Morgan, J. T. McNamara, M. Fischer, J. Rich, H. M. Chen, S. G. Withers, J. Zimmer, Observing cellulose biosynthesis and membrane translocation in crystallo. *Nature* **531**, 329–334 (2016).
 30. J. F. Acheson, R. Ho, N. F. Goularte, L. Cegelski, J. Zimmer, Molecular organization of the *E. coli* cellulose synthase macrocomplex. *Nat. Struct. Mol. Biol.* **28**, 310–318 (2021).
 31. L. L. Lairson, B. Henrissat, G. J. Davies, S. G. Withers, Glycosyltransferases: Structures, functions, and mechanisms. *Annu. Rev. Biochem.* **77**, 521–555 (2008).
 32. V. Malik, G. W. Black, Structural, functional, and mutagenesis studies of UDP-glycosyltransferases. *Adv. Protein Chem. Struct. Biol.* **87**, 87–115 (2012).
 33. S. J. Charnock, G. J. Davies, Structure of the nucleotide-diphospho-sugar transferase, SpsA from *Bacillus subtilis*, in native and nucleotide-complexed forms. *Biochemistry* **38**, 6380–6385 (1999).
 34. A. Punjani, J. L. Rubinstein, D. J. Fleet, M. A. Brubaker, cryoSPARC: Algorithms for rapid unsupervised cryo-EM structure determination. *Nat. Methods* **14**, 290–296 (2017).
 35. J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Židek, A. Potapenko, A. Bridgland, C. Meyer, S. A. A. Kohl, A. J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A. W. Senior, K. Kavukcuoglu, P. Kohli, D. Hassabis, Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
 36. M. Varadi, S. Anyango, M. Deshpande, S. Nair, C. Natassia, G. Yordanova, D. Yuan, O. Stroe, G. Wood, A. Laydon, A. Židek, T. Green, K. Tunyasuvunakool, S. Petersen, J. Jumper, E. Clancy, R. Green, A. Vora, M. Lutfi, M. Figurnov, A. Cowie, N. Hobbs, P. Kohli, G. Kleywegt, E. Birney, D. Hassabis, S. Velankar, AlphaFold Protein Structure Database: Massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Res.* **50**, D439–D444 (2022).
 37. C. Jimenez-Ortigosa, J. Jiang, M. Chen, X. Kuang, K. R. Healey, P. Castellano, N. Boparai, S. J. Ludtke, D. S. Perlin, W. Dai, Preliminary structural elucidation of β -(1,3)-glucan synthase from *Candida glabrata* using cryo-electron tomography. *J. Fungi (Basel)* **7**, 120 (2021).
 38. P. Zhao, C. Zhao, D. Chen, C. Yun, H. Li, L. Bai, Structure and activation mechanism of the hexameric plasma membrane H⁺-ATPase. *Nat. Commun.* **12**, 6439 (2021).
 39. X. Hu, P. Yang, C. Chai, J. Liu, H. Sun, Y. Wu, M. Zhang, M. Zhang, X. Liu, H. Yu, Structural and mechanistic insights into fungal β -1,3-glucan synthase FKS1. *Nature* **616**, 190–198 (2023).
 40. Z. Ren, A. Chhetri, Z. Guan, Y. Suo, K. Yokoyama, S. Y. Lee, Structural basis for inhibition and regulation of a chitin synthase from *Candida albicans*. *Nat. Struct. Mol. Biol.* **29**, 653–664 (2022).
 41. F. P. Maloney, J. Kuklewicz, R. A. Corey, Y. Bi, R. Ho, L. Mateusiak, E. Pardon, J. Steyaert, P. J. Stansfeld, J. Zimmer, Structure, substrate recognition and initiation of hyaluronan synthase. *Nature* **604**, 195–201 (2022).
 42. S. Q. Zheng, E. Palovcak, J. P. Armache, K. A. Verba, Y. Cheng, D. A. Agard, MotionCor2: Anisotropic correction of beam-induced motion for improved cryo-electron microscopy. *Nat. Methods* **14**, 331–332 (2017).
 43. J. A. Mindell, N. Grigorieff, Accurate determination of local defocus and specimen tilt in electron microscopy. *J. Struct. Biol.* **142**, 334–347 (2003).
 44. J. Zivanov, T. Nakane, B. O. Forsberg, D. Kimanius, W. J. H. Hagen, E. Lindahl, S. H. W. Scheres, New tools for automated high-resolution cryo-EM structure determination in RELION-3. *eLife* **7**, e24166 (2018).
 45. P. Emsley, B. Lohkamp, W. G. Scott, K. Cowtan, Features and development of Coot. *Acta Crystallogr. D Biol. Crystallogr.* **66**, 486–501 (2010).
 46. E. F. Pettersen, T. D. Goddard, C. C. Huang, E. C. Meng, G. S. Couch, T. I. Croll, J. H. Morris, T. E. Ferrin, UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Sci.* **30**, 70–82 (2021).
 47. P. D. Adams, P. V. Afonine, G. Bunkóczi, V. B. Chen, I. W. Davis, N. Echols, J. J. Headd, L. W. Hung, G. J. Kapral, R. W. Grosse-Kunstleve, A. J. McCoy, N. W. Moriarty, R. Oeffner, R. J. Read, D. C. Richardson, J. S. Richardson, T. C. Terwilliger, P. H. Zwart, PHENIX: A comprehensive Python-based system for macromolecular structure solution. *Acta Crystallogr. D Biol. Crystallogr.* **66**, 213–221 (2010).
 48. V. B. Chen, W. B. Arendall III, J. J. Headd, D. A. Keedy, R. M. Immormino, G. J. Kapral, L. W. Murray, J. S. Richardson, D. C. Richardson, MolProbity: All-atom structure validation for macromolecular crystallography. *Acta Crystallogr. D Biol. Crystallogr.* **66**, 12–21 (2010).

Acknowledgments: Cryo-EM data were collected in the Electron Microscopy Laboratory of Peking University and Cryo-EM Platform at the Center for Biological Imaging (CBI; cbi.ibp.ac.cn) at the Institute of Biophysics, Chinese Academy of Sciences. We thank Xuemei Li, Z. Guo, B. Zhu, Xujing Li, and X. Huang for facilitating data collection. We thank H. Deng and X. Wang in Institute of Biophysics for help in construct preparation. We thank L. Bianji (Edanz) for editing the English text of the draft of this manuscript. **Funding:** This work was supported by grants from the National Natural Science Foundation of China (32171212 to L.B., 32071207 to C.-H.Y., and 82071658 to J.H.), the Fundamental Research Funds for the Central Universities (to L.B.), the National Key R&D Program of China (2022YFC2702904 to J.H.), the National Basic Research Program of China (973 Program, 2012CB917202 to C.-H.Y.), and the Beijing Nova Program (Z201100006820010 and 20220484160 to J.H.). **Author contributions:** C.-R.Z., Z.-L.Y., J.H., J.Q., C.-H.Y., and L.B. conceived and designed the experiments. C.-R.Z., Z.-L.Y., D.-D.C., Z.-B.W., J.H., M.J., L.-X.W., P.Z., and L.B. performed the experiments. C.-R.Z., D.-D.C., Z.-L.Y., J.H., J.Q., C.-H.Y., and L.B. analyzed the data. J.H., J.Q., C.-H.Y., and L.B. wrote the manuscript with input from all authors. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** The cryo-EM 3D map and the corresponding atomic model of the Fks1 have been deposited at the EMDB (<https://ebi.ac.uk/emdb/>) database and the RCSB PDB (<https://rcsb.org/>) with the respective accession codes of EMD-36748 and 8JZN. All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials.

Submitted 14 March 2023
Accepted 10 August 2023
Published 13 September 2023
10.1126/sciadv.adh7820